

Kubernetes · SSO

# Kubernetes Cluster SSO Setup

How we unified ArgoCD, Grafana, Argo Workflows, Airflow, and Kubecost under a single Google account using Authentik

Jonghwa Baek



## Problem

# Separate accounts everywhere

Every new joiner needed accounts created on each service. Every departure meant hunting down and removing them one by one.

**ArgoCD**

Separate account

**Grafana**

Separate account

**Argo Workflows**

Separate account

**Airflow**

Separate account

**Kubecost**

No auth (IP restriction only)

Why Authentik?

## Choosing from 3 candidates

Keycloak is UI-click centric, making it hard to manage with GitOps. Authentik lets you declare app config, policies, and groups as code via Blueprint YAML.

**Okta**

SaaS · Per-user pricing

**Keycloak**

Self-hosted · UI-centric

**Authentik**

Self-hosted · Blueprint YAML

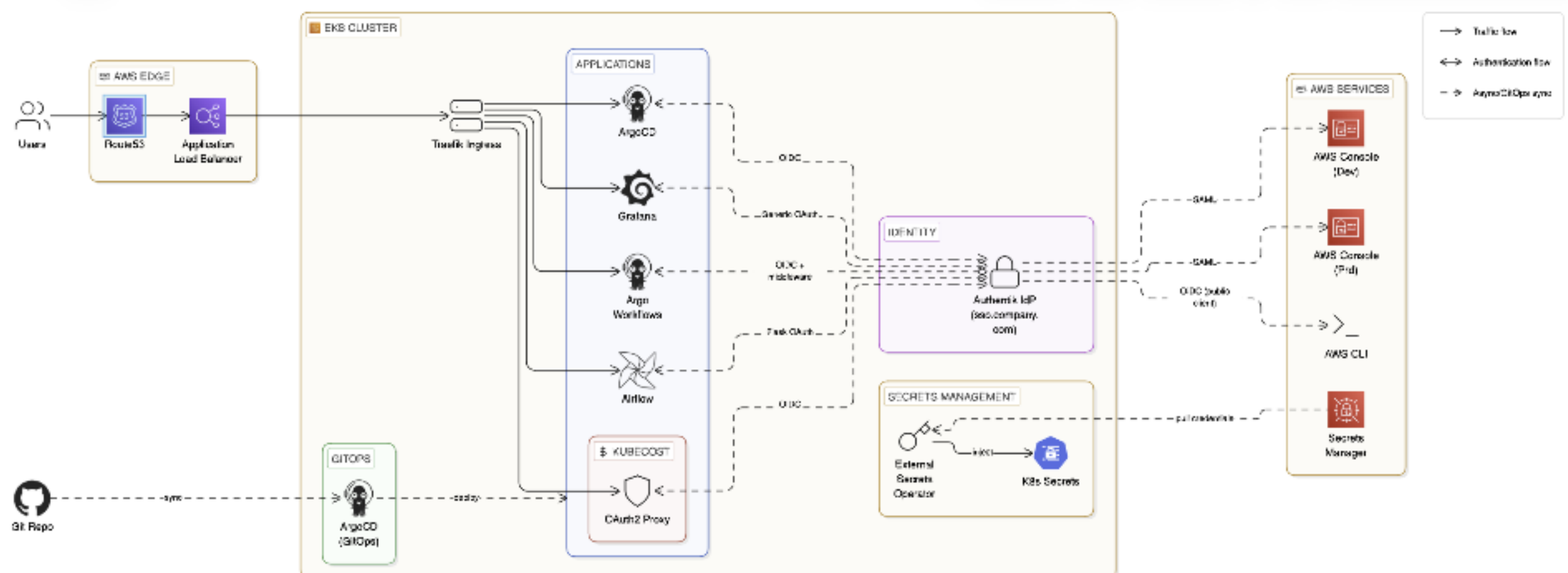
**SELECT**

# Full Flow

Requests route through Route53 - ALB - Traefik to each service.

Each service delegates auth to Authentik via OIDC/OAuth2/SAML.

OAuth secrets are auto-injected from AWS Secrets Manager into K8s Secrets via External Secrets.



## 4 Authentication Patterns

Each app supports different protocols — there's no one-size-fits-all. You pick the right pattern per case.

### OIDC

**ArgoCD, Argo Workflows**

Public Client + PKCE — same client for web and CLI login

### Generic OAuth

**Grafana**

Auto role mapping via `role_attribute_path` based on groups

### oauth2-proxy

**Kubecost**

Proxy placed in front of apps with no built-in auth

### SAML

**AWS Console**

Authentik as IdP — temporary credentials, no IAM Users

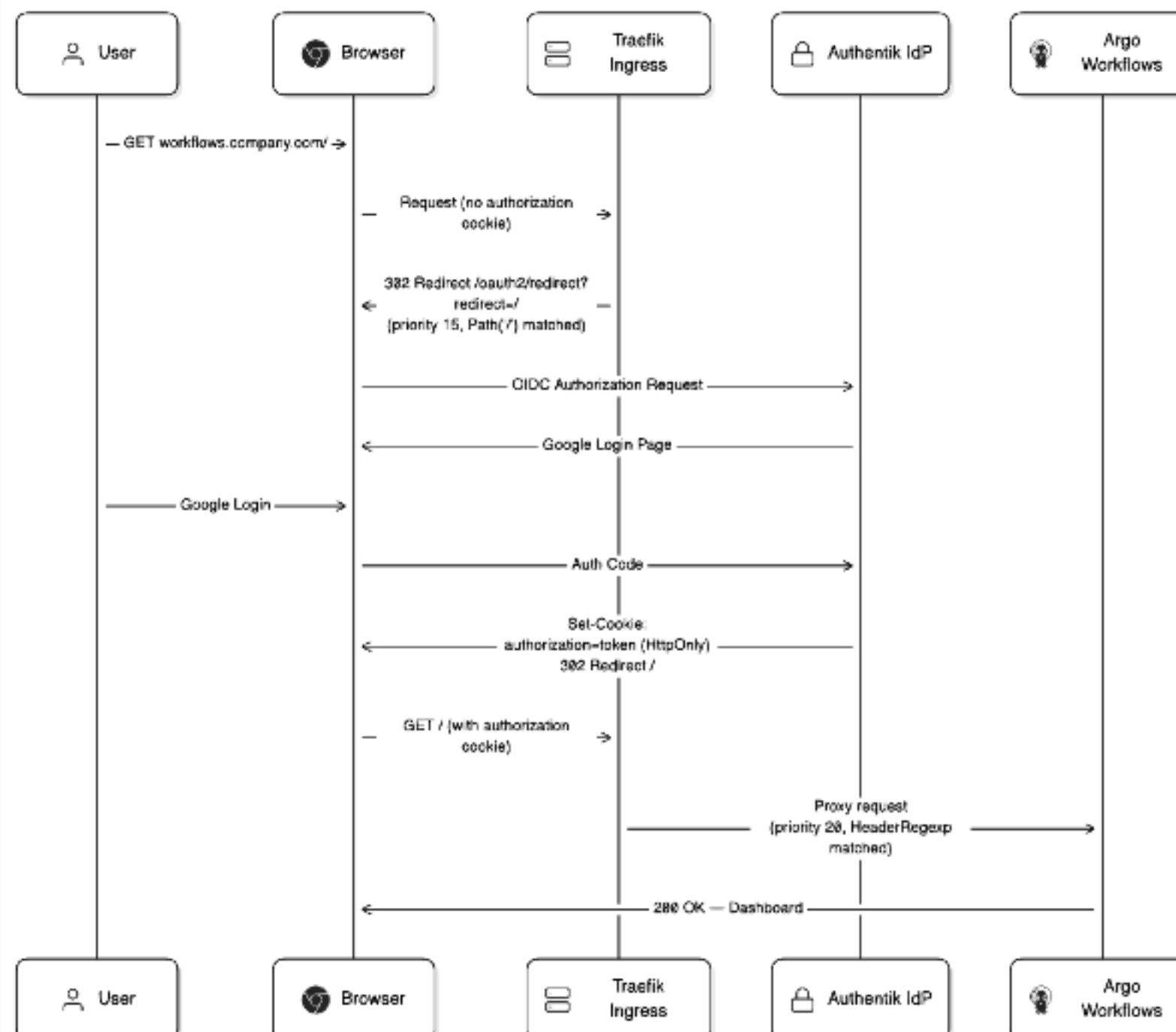


# Argo Workflows Infinite Redirect

We added an SSO redirect middleware in Traefik, but after login it kept redirecting. Fixed by routing on the presence of the authorization cookie set after SSO callback. Compounded by Traefik v3 renaming HeadersRegexp to HeaderRegexp (singular).

Before: / - SSO - done - / - SSO again - loop

After: cookie present = pass through, missing = redirect to SSO



# Before vs After

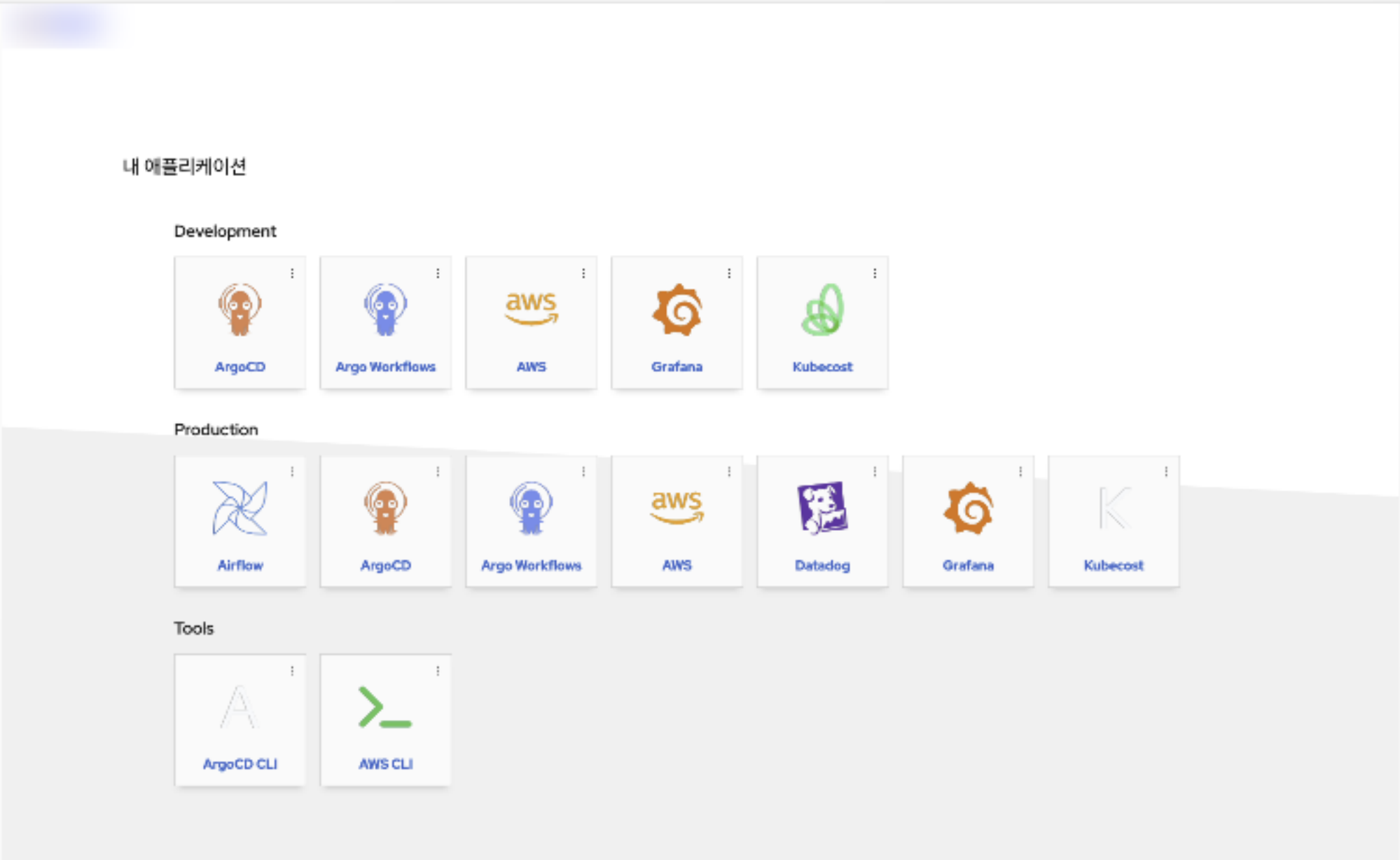
The biggest win was security. Long-lived IAM User credentials are fully removed — AWS Console access now uses SAML temporary credentials only.

Before

- Manually create accounts per service
- Manually delete accounts per service
- IAM User (long-lived credentials)
- Manually issue access keys

After

- Google account + add to group once
- Disable Authentik account once
- SAML (temporary credentials)
- OIDC auto-refresh



## Wrap-up

"Just plug in an SSO — how hard can it be?" Then every app had a different protocol, K8s version changes, and a Traefik v3 migration all hit at once.

- Adding a new service = one app entry in the Blueprint
- Long-lived credentials (IAM User) fully eliminated
- SSO config managed as code via GitOps

[b100to.github.io](https://b100to.github.io)